Egypt Environmental Monitor

An end-to-end data pipeline for air quality, weather and earthquake monitoring in Egypt

Project Documentation: Parts 1-3

Project Planning & Management | Literature Review | Requirements Gathering

Initiative	Digital Egypt Pioneers Initiative (DEPI), Round 5
Track	Data Engineering
Team / Group	Egypt Environmental Monitor
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Part 1: Project Planning & Management

1.1 Project idea and problem statement

Egypt faces recurring environmental and natural-hazard concerns: seasonal air pollution episodes (for example the autumn "black cloud" and dust storms), and seismic activity around the Gulf of Suez, the Gulf of Aqaba and the Mediterranean. Data about these topics exists, but it is spread across different providers, in different formats and units, and is rarely presented together in one place with historical context.

Egypt Environmental Monitor is a data engineering project that collects air quality, weather and earthquake data from open sources, streams it through a message broker, cleans and models it in a layered (Bronze / Silver / Gold) architecture, and exposes it through dashboards and automatic alerts. The three domains are combined in a single pipeline because they share the same pattern (ingest, store, stream, model, serve) and because weather helps explain pollution levels.

1.2 Objectives

- Ingest near-real-time air quality (PM2.5, PM10, NO2, O3, AQI) and weather (temperature, wind speed, humidity) data for 6 Egyptian cities.
- Ingest live and historical earthquake data for Egypt and the surrounding region.
- Build a reliable, automated pipeline: Kafka-compatible streaming, historical backfill, Bronze to Silver to Gold transformation, orchestrated by Airflow.
- Deliver at least 4 tested Gold tables (dbt) that answer concrete analytical questions.
- Provide dashboards and at least one working alert (air quality threshold or earthquake magnitude threshold).
- Deliver a reproducible project: the whole stack starts with `docker compose up`, with documentation and a data dictionary.

1.3 Scope

Cities: Cairo, Giza, Alexandria, Mansoura, Aswan, Sharm El-Sheikh. **Earthquake area:** an approximate bounding box around Egypt and the surrounding region (starting proposal: latitude 20 to 35, longitude 24 to 38), to be tuned after the first experiments.

In scope (MVP)	Out of scope / stretch goals
3+ cities with air quality and weather data	Spark processing (only if time allows)
Earthquake feed for Egypt and the region	NASA FIRMS fire data
Historical backfill (weather + earthquakes)	Data lineage tooling
4 Gold tables built with dbt and tested	Simple forecasting model
1+ Airflow DAG	Real WebSocket stream from EMSC (optional)
1 dashboard and 1 working alert	Mobile app or public web portal
Setup through Docker Compose	Health-risk medical advice

1.4 Solution architecture

Data flows from the public APIs through Python pollers into Kafka-compatible topics (Redpanda). A stream consumer writes raw data to the Bronze layer. Historical backfill jobs, orchestrated by Airflow, land into the same layer. Cleaning, de-duplication and typing produce Silver; dbt models and tests produce Gold; Grafana or Metabase serves dashboards and alerts.

Layer	What happens	Technology
Sources	Air quality and weather (Open-Meteo), live earthquakes (USGS GeoJSON feed), historical earthquakes (USGS FDSN catalog), optional EMSC WebSocket	REST APIs, GeoJSON
Ingestion	Python pollers run every 1 to 15 minutes and publish to topics `air_quality_raw`, `weather_raw`, `earthquakes_raw`	Python, Redpanda (Kafka API)
Bronze	Raw, immutable data stored as received	Parquet
Silver	Cleaning, de-duplication, unit and timestamp standardization, typing, basic validity checks	Python consumers, pandas / DuckDB, Parquet / Postgres
Gold	Analytical models with tests	dbt, Postgres / TimescaleDB
Serving	Dashboards, Pipeline Health page, alerts (Telegram / Email)	Grafana (or Metabase)
Orchestration & DevOps	Scheduling of backfill and dbt; containerized setup; CI	Airflow, Docker Compose, Git, GitHub Actions

Planned Gold tables

Table	Question it answers
city_air_quality_hourly	What is the air quality in each city, hour by hour?
air_quality_vs_baseline	Is today's air worse than usual for the same month?
weather_pollution_correlation	How do wind, temperature and humidity relate to PM2.5?
earthquakes_by_region_month	How many earthquakes, and how strong, by region and month?
earthquake_activity_vs_baseline	Is current seismic activity above the historical average?
active_alerts	Which alerts are active now (poor air quality or an earthquake above a threshold)?

1.5 Project plan (12 weeks)

Week	Goal	Deliverable
1-2	Learn/refresh Git, Docker, Python, SQL; understand the APIs	Repository ready; first successful API request
3	Docker Compose (Redpanda + Postgres); first air quality poller	Events arriving in Kafka
4	Weather and earthquake pollers; Bronze storage	3 topics running
5	Historical backfill for weather and earthquakes	Historical data in Bronze
6	Silver: cleaning, de-duplication, timestamp and unit standardization	Clean Silver tables
7-8	Gold models with dbt and tests	4 to 6 tested Gold tables
9	Airflow: orchestrate backfill and dbt	Automatic pipeline
10	Data quality and monitoring (freshness, job failures, Pipeline Health page)	Quality checks and health view
11	Dashboards and alerts (Email or Telegram)	Working dashboard and alert
12	Documentation, architecture diagram, presentation rehearsal	Project ready for defense

Note: if the remaining time before the final defense is shorter than 12 weeks, the scope will be reduced (for example 3 cities instead of 6, with alerts postponed) rather than compressing quality work. Replace the week numbers with real

calendar dates once the team fixes its start date.

1.6 Team structure and responsibilities

#	Role	Responsibilities	Owner
1	Air Quality & Weather Ingestion	Pollers for air quality and weather, Kafka topics, rate-limit handling	Beshoy Magdy Farid
2	Earthquake Ingestion	USGS poller (optional EMSC WebSocket), bounding-box filtering	Manar Mousa Hossan
3	Historical Backfill	Historical weather and earthquake loading, conversion to Parquet	Mohamed Gamal Ahmed
4	Transformation & Modeling	Silver and Gold layers, dbt models, schema design	Mohamed Reda Mohamed
5	DevOps, Quality & Dashboards	Docker, Airflow, CI, data quality, dashboards, alerts	Mohamed Hossam Ahmed

Collaboration rules: one GitHub repository for the team; feature branches and pull requests with at least one reviewer; short weekly sync and a task board; roles are rotated or paired every 3 to 4 weeks so that every member works on more than one layer of the pipeline.

1.7 Risk management

Risk	Likelihood / Impact	Mitigation
API terms, limits or availability change (Open-Meteo, USGS)	Medium / High	Review terms before starting; cache and back off on errors; Replay mode with a recorded sample so the demo works offline
Air quality data is model-based (CAMS), not ground sensors, and coarse for Egypt	High / Medium	State clearly in the documentation and dashboard; optionally cross-check with OpenAQ stations where they exist
Shorter history for air quality than for weather	High / Medium	Check the documented history window before designing the backfill; design baselines accordingly
Weak laptops cannot run the full stack	Medium / Medium	Use DuckDB instead of Spark; run Airflow on one machine only; limit the number of containers
Team members unfamiliar with Kafka, dbt or Airflow	High / Medium	Learning weeks at the start; pair programming; role rotation
Scope too large for the time left	Medium / High	Strict MVP; stretch goals only after the MVP is stable
Integration problems between layers	Medium / Medium	Agree on schemas early; data dictionary; CI checks; frequent integration instead of a last-week merge
A team member becomes unavailable	Low / Medium	Pairing, documentation of every component, shared repository

1.8 Success indicators (KPIs)

Indicator	Target
Cities with air quality and weather data flowing	>= 3 (goal: 6)
Historical backfill completed (weather and earthquakes)	Yes
Gold tables built with dbt and passing tests	>= 4
Airflow DAGs running successfully	>= 1, with successful scheduled runs
Data freshness (age of the newest record per source)	Within a few polling intervals; monitored on the Pipeline Health page
Dashboards and alerts	>= 1 dashboard and >= 1 alert that fires correctly in a test
Reproducibility	A new machine can start the stack with `docker compose up` following the README
Documentation	Architecture diagram, data dictionary, README and final presentation completed

1.9 Repository structure

```
egypt-env-monitor/
  docker-compose.yml, README.md
  ingestion/ (air_weather_poller/, earthquake_poller/)
  backfill/ (historical loading)
  processing/ (Bronze to Silver)
  dbt/ (Gold models and tests)
  airflow/dags/
  dashboards/
  docs/ (architecture and data dictionary)
  .github/workflows/
```

Part 2: Literature Review

2.1 Method

We looked for existing public services, datasets and national institutions that monitor air quality, weather or earthquakes in Egypt, and for the data providers our pipeline depends on. For each we recorded what it offers, its strengths and limitations, and how our project differs. Sources are listed in section 2.6. The review should be extended by the team with academic papers on Egyptian air pollution before the final submission if the track requires it.

2.2 Existing solutions

A. IQAir (AirVisual): public air quality platform

IQAir publishes real-time AQI and PM2.5 pages for Egyptian locations such as Cairo and New Cairo, with forecasts, station rankings and historical graphs. **Strengths:** polished consumer interface, station-based data, forecasts. **Limitations:** a closed consumer product; it does not combine weather correlation or earthquakes, offers no transparent data pipeline or reproducible datasets for analysis, and full data access is a commercial offering.

B. OpenAQ: open air quality data platform

OpenAQ aggregates measurements from ground monitoring stations worldwide and exposes them through an API (a free API key is required). **Strengths:** measured data, open and programmatic. **Limitations:** coverage in Egypt is limited, so availability must be verified for each chosen city. We treat it as an optional validation source rather than the primary feed.

C. Open-Meteo Air Quality API (CAMS-based)

Open-Meteo republishes modeled data from the Copernicus Atmosphere Monitoring Service (CAMS) with PM2.5, PM10, NO2, O3 and AQI values, no API key, and hourly resolution. **Strengths:** free for non-commercial use, simple API, global coverage, historical and forecast ranges. **Limitations:** values are model output, not sensor measurements. The high-resolution (about 11 km) CAMS European model covers Europe only; for locations like Egypt the coarser global model (about 40 to 45 km) applies, so neighboring cities such as Cairo and Giza may fall in the same grid cell. The documented history for air quality is also shorter than for weather. Attribution to CAMS and Open-Meteo is required.

D. Egyptian National Seismological Network (ENSN, NRIAG)

ENSN is Egypt's official seismic network, operated by the National Research Institute of Astronomy and Geophysics (NRIAG). It started operating in 1997 after the 1992 Cairo earthquake (Mw 5.8), with its main center in Helwan, Cairo and regional sub-centers; it publishes earthquake information and annual bulletins. **Strengths:** the authoritative local source with dense national coverage. **Limitations:** oriented to scientific and official reporting rather than to a ready-to-use analytical data pipeline, and programmatic access for a student project is not as straightforward as with USGS.

E. USGS and EMSC earthquake services

USGS provides GeoJSON feeds that are updated roughly every minute and a searchable historical catalog (FDSN web service), which makes it suitable for filtering by bounding box and time. EMSC offers a WebSocket for genuinely streaming earthquake notifications. **Strengths:** free, well-documented, global. **Limitations:** global agencies may report fewer small local events than the national network, and the feeds are polled, not pushed (except EMSC).

2.3 Comparison with our project

Solution	Air quality	Weather	Earthquakes	Open pipeline / history	Alerts
IQAir	Yes (stations)	Limited	No	No (closed product)	App-based
OpenAQ	Yes (stations, limited in Egypt)	No	No	API; no analytics layer	No
Open-Meteo	Yes (modeled)	Yes	No	API only; no modeling layer	No
ENSN / NRIAG	No	No	Yes (official)	Bulletins / web	Official channels
USGS / EMSC	No	No	Yes (global)	Feeds and catalog	Notification services
Egypt Environmental Monitor	Yes (modeled; optional station check)	Yes	Yes	**Full pipeline: streaming, backfill, Bronze/Silver/Gold, tests**	**Yes (Telegram / Email)**

2.4 Gap and contribution

- Existing tools each cover a single domain; none we found combines air quality, weather and earthquakes for Egypt in one place with analytical context.
- Most services show the current value only. Our Gold layer compares today with a monthly baseline (air quality) and with the historical average (earthquake activity).
- The project explains pollution using weather (wind, temperature, humidity) through a correlation table.
- The pipeline is open, reproducible and tested, and includes data quality checks and a Pipeline Health view, which consumer apps do not expose.

2.5 Lessons applied to our design

- Be explicit that Open-Meteo air quality is modeled data and show this on the dashboard.
- Check OpenAQ coverage per city before relying on it; keep it optional.
- Check the available history window of each API before designing the backfill and baselines.
- Treat polling as near-real-time streaming through Kafka and provide a Replay mode so demonstrations do not depend on live internet.
- Detect impossible sensor/model values (for example negative PM2.5) as a visible data quality feature.

2.6 References

[1] IQAir. Air quality in Cairo / New Cairo (real-time AQI and PM2.5, station ranking). https://www.iqair.com/egypt/cairo (accessed October 2026).

[2] Open-Meteo. Air Quality API documentation (CAMS European and global domains, resolution and availability). https://open-meteo.com/en/docs/air-quality-api

[3] Open-Meteo. Air quality API now available (announcement; CAMS-based PM2.5, PM10, dust and gases). https://openmeteo.substack.com/p/air-quality-api-now-available

[4] ECMWF / Copernicus Atmosphere Monitoring Service. CAMS European air quality forecasts (0.1 degree, ensemble of eleven models). https://www.ecmwf.int/node/37291

[5] FDSN. EY: Egyptian National Seismological Network (ENSN), operated by NRIAG, since 1997. https://fdsn.org/networks/detail/EY

[6] NRIAG. Egyptian National Seismic Network (ENSN) Lab. https://www.nriag.sci.eg/?p=423

[7] NRIAG. Egyptian Earthquakes Annual Bulletin 2021. https://www.nriag.sci.eg/wp-content/uploads/2023/03/Earthquakes-Annual-Bulletin-2021.pdf

[8] AbdelHafiez, H. E. and Tawfiek, A. (2020). A review of the Egyptian National Seismological Network after 20 years of operation. NRIAG Journal of Astronomy and Geophysics, 9(1), 330-340.

[9] USGS Earthquake Hazards Program. Real-time GeoJSON feeds and the FDSN Event Web Service (earthquake catalog). https://earthquake.usgs.gov/

[10] OpenAQ. Open air quality data platform and API. https://openaq.org/

[11] EMSC (European-Mediterranean Seismological Centre). Real-time earthquake information and WebSocket service. https://www.emsc-csem.org/

Part 3: Requirements Gathering

3.1 Stakeholders and their needs

Stakeholder	Description	Main needs
General public / residents	People in the six cities, including those sensitive to air pollution (asthma, allergies)	Easy-to-read current air quality and weather; alerts when air becomes unhealthy
Environmental analysts and researchers	Students, researchers, NGOs	Clean historical datasets, baseline comparisons, weather-pollution relationships
Seismic-interest users	Engineers, students, interested public	Recent earthquakes around Egypt, monthly counts and magnitudes, comparison with the historical average
Decision makers / local authorities	Municipal or environmental officials (potential users)	Trends and summaries of pollution episodes; evidence for planning
Project team and instructors	Developers, maintainers, evaluators	Reproducible setup, clear documentation, monitoring of pipeline health, quality evidence

3.2 Functional requirements

ID	Requirement	Priority
FR-01	The system shall poll air quality (PM2.5, PM10, NO2, O3, AQI) for each configured city at a configurable interval and publish it to the `air_quality_raw` topic.	Must
FR-02	The system shall poll weather (temperature, wind speed, humidity) for each configured city and publish it to the `weather_raw` topic.	Must
FR-03	The system shall poll the USGS earthquake feed, filter events by the configured bounding box, and publish them to the `earthquakes_raw` topic.	Must
FR-04	The system shall store all raw data in the Bronze layer without modification.	Must
FR-05	The system shall provide a historical backfill for weather and earthquakes (and for air quality within the history the API offers).	Must
FR-06	The system shall clean, de-duplicate, type, and standardize timestamps and units in the Silver layer.	Must
FR-07	The system shall build at least four Gold tables with dbt: hourly city air quality, air quality vs baseline, earthquakes by region and month, and an alerts table (plus weather-pollution correlation and earthquake baseline as planned).	Must
FR-08	The system shall run dbt tests on Gold models and fail visibly when tests fail.	Must
FR-09	The system shall orchestrate backfill and transformation with at least one Airflow DAG.	Must
FR-10	The system shall provide a dashboard with air quality by city, weather, earthquake map/summary and active alerts.	Must
FR-11	The system shall send at least one type of alert (Email or Telegram) when air quality exceeds a configured threshold or an earthquake exceeds a configured magnitude. Proposed defaults: US AQI above 100 for air quality; magnitude threshold to be agreed by the team.	Must
FR-12	The system shall detect impossible or suspicious values (for example negative PM2.5) and flag them in the data quality checks.	Should
FR-13	The system shall provide a Pipeline Health view showing data freshness and the status of recent jobs.	Should
FR-14	The system shall provide a Replay mode that re-plays a recorded sample of events through Kafka for offline demonstrations.	Should
FR-15	The system shall optionally ingest EMSC WebSocket events and cross-check OpenAQ station data where available.	Could

3.3 Non-functional requirements

Category	Requirement
Performance	Pollers shall complete each polling cycle well inside their interval; the dashboard shall load its main views in a few seconds on the target data volume; the full stack shall run on a typical student laptop (DuckDB used instead of Spark where needed).
Reliability	Pollers shall handle API errors, timeouts and rate limits with retries and back-off without crashing; failed Airflow tasks shall be retried and reported.
Data quality	Freshness, range and uniqueness checks on Silver and Gold; de-duplication so that re-running jobs does not create duplicate records (idempotency).
Security	API keys, tokens and passwords shall be stored in environment variables or secret files and never committed to GitHub; dashboards and databases use non-default credentials.
Usability	Dashboards shall be readable by non-technical users, show units and time zone clearly, and state that air quality values come from a model (CAMS).
Maintainability	Modular code organized as in the repository structure, formatted and checked in CI (GitHub Actions); README and data dictionary kept up to date.
Portability / reproducibility	The stack shall start with `docker compose up` from a clean checkout; versions of images and libraries are pinned.
Compliance	Respect each provider's terms and rate limits; keep the required attribution to CAMS and Open-Meteo; use data for non-commercial (graduation project) purposes.
Scalability	Adding a new city or changing the bounding box shall require configuration changes only, not code changes.

3.4 Data requirements

Data	Source	Frequency	Notes
Air quality (PM2.5, PM10, NO2, O3, AQI)	Open-Meteo Air Quality API	Polled every 1 to 15 min (model updates less often)	Modeled CAMS data; no key; attribution required
Weather (live and history)	Open-Meteo Forecast and Historical APIs	Polled; long history	No key; non-commercial use
Earthquakes (live)	USGS GeoJSON feeds	About every minute	Filter by bounding box
Earthquakes (history)	USGS FDSN catalog	Backfill by period	Bounding box and time filters
Station air quality (optional)	OpenAQ	As available	Free API key; limited coverage in Egypt

3.5 Assumptions and constraints

- Sources are free and public; their terms, limits and availability may change, so they are reviewed before and during development.
- The project is for educational, non-commercial use.
- Near-real-time polling is treated as a stream through Kafka; true push streaming exists only if the optional EMSC WebSocket is added.
- Air quality values are modeled estimates and are not official health advisories.
- Team of five members; limited time and student hardware; the schedule follows the graduation project deadlines of the initiative.

3.6 Acceptance criteria (aligned with the MVP)

#	Criterion	Verified by
1	Air quality and weather data flows for at least 3 cities	Rows present in Bronze/Silver for each city; dashboard
2	Earthquakes ingested for Egypt and the region	Events present in topic and tables
3	Historical backfill for weather and earthquakes completed	Date-range check on Bronze/Silver
4	Four Gold tables built with dbt and tests passing	dbt build output / CI
5	At least one Airflow DAG runs successfully	Airflow UI run history
6	One dashboard and one alert working	Live demo or recorded Replay-mode demo; alert test message
7	Whole stack starts with docker compose up	Fresh-machine test following the README

3.7 Example user stories

- As a resident of Cairo, I want to see whether PM2.5 is above normal today, so that I can decide whether to limit outdoor activity.
- As an analyst, I want to see how wind speed relates to PM2.5 in each city, so that I can explain pollution episodes.
- As a student of seismology, I want monthly earthquake counts and magnitudes by region compared with the historical average, so that I can see unusual periods.
- As a team member, I want a Pipeline Health page, so that I can see immediately if data is stale or a job has failed.